Lecture 7

Lattices, centres and global forms

Objectives. *The groups* $\mathrm{SU}(2)$ *and* $\mathrm{SO}(3)$ *have isomorphic Lie algebras but are not isomorphic. This lecture asks which discrete data determine connected groups with the same Lie algebra. For compact groups, the answer is the kernel lattice* Λ *of a maximal torus, inside its Lie algebra* $\mathfrak{t}$*, read together with the root data.*

The lecture proves this for semisimple Lie algebras, using cited structure results. Different lattices can still give isomorphic groups, through automorphisms of the Lie algebra that preserve $\mathfrak{t}$*. When there is a central torus, the lecture shows how* Λ *determines a finite group* F*.*

By the end of this lecture you will be able to:

- Prove that a compact connected abelian Lie group is $\mathfrak{t}/\Lambda$, with characters the ℓ satisfying $\ell(\Lambda) \subset 2\pi\mathbb{Z}$.
- Place Λ between the coroot and coweight lattices, and compute centres.
- Describe the connected compact groups (global forms) with a given semisimple Lie algebra $\mathfrak{k}$, and the groups $(T_0 \times K_{sc})/F$.
- Decide which representations descend to $\mathrm{SO}(3)$, $\mathrm{PSU}(3)$ and $\mathrm{U}(2)$.

The Lie groups denoted K or T in this lecture, with or without subscripts, are connected. The subgroups C, F and $Z(K)$ that appear later need not be connected. A Lie group and its identity component have the same Lie algebra, so we restrict to connected groups. Even connected groups with isomorphic Lie algebras need not be isomorphic, as $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ show.

7.1 Compact connected abelian groups

Definition 7.1. A *torus* is a Lie group isomorphic to $\mathrm{U}(1)^n$ for some $n \geq 0$. A *lattice* in a real vector space V is a subgroup $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_n$, where $v_1, \ldots, v_n$ is a basis of V.

Lemma 7.2 (Discrete subgroups)**.** *Every discrete subgroup of a finite-dimensional real vector space equals* $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_k$ *for some linearly independent* $v_1, \ldots, v_k$*.*

We cite this standard lemma from BD, Section I.3, Lemma (3.8).

Theorem 7.3. *Let* T *be a compact connected abelian Lie group with Lie algebra* $\mathfrak{t}$ *of dimension* n*. Then* $\exp\colon \mathfrak{t} \to T$ *is a surjective homomorphism, and* $\Lambda = \ker \exp$ *is a lattice*

in $\mathfrak{t}$. *For a basis* $v_1, \dots, v_n$ *of* Λ, *the map*

$$\Phi\colon \mathrm{U}(1)^n \to T, \qquad \Phi\big(e^{2\pi i x_1}, \dots, e^{2\pi i x_n}\big) = \exp(x_1 v_1 + \cdots + x_n v_n),$$

is an isomorphism of Lie groups.

Proof. Since T is abelian, so is $\mathfrak{t}$. Commuting elements satisfy $\exp(X+Y) = \exp X \exp Y$ (Lecture 2), so exp is a homomorphism. Choose an open set $U = -U$ containing 0 that exp maps diffeomorphically onto an open set (Lecture 2).

Surjectivity. The subgroup $\exp(\mathfrak{t})$ contains $g\exp(U)$ for each of its elements g, so it is open. Its complement is a union of open cosets, so it is also closed. Since T is connected, $\exp(\mathfrak{t}) = T$.

Discreteness. As exp is injective on U, we have $\Lambda \cap U = \{0\}$. If $H, H' \in \Lambda$ and $H' \in H + U$, then $H' - H \in \Lambda \cap U$, so $H' = H$. Hence Λ is discrete. The discrete-subgroup lemma, Lemma 7.2, gives independent generators $v_1, \dots, v_k$ of Λ.

Full rank. Suppose $k < n$. Choose a nonzero linear map $p\colon \mathfrak{t} \to \mathbb{R}$ with $p(v_1) = \cdots = p(v_k) = 0$, and define

$$f\colon T \to \mathbb{R}, \qquad f(\exp H) = p(H).$$

It is well defined because $p(\Lambda) = 0$. It is continuous: near $\exp H_1$ we have $f(\exp H_1 \exp Y) = p(H_1) + p(Y)$ for $Y \in U$, and $Y \mapsto \exp H_1 \exp Y$ is a homeomorphism of U onto a neighbourhood of $\exp H_1$. So $\mathbb{R} = p(\mathfrak{t}) = f(T)$ is compact, which is false. Hence $k = n$.

Isomorphism. Changing x_j by an integer changes $\sum_j x_j v_j$ by an element of Λ. So Φ is a well-defined homomorphism. It is continuous, because near any point it is the composite of a local inverse of $x \mapsto (e^{2\pi i x_j})_j$ with the continuous map $x \mapsto \exp(\sum_j x_j v_j)$. It is bijective because $\ker\exp = \bigoplus_j \mathbb{Z} v_j$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. Continuous homomorphisms of Lie groups are smooth (Lecture 3), so Φ and Φ^{-1} are smooth. ∎

Lecture 6 defined a torus as a compact connected abelian Lie group. The theorem shows that such a group is a torus in the sense above, and the converse is clear. So the two definitions agree. The theorem also proves the statement $T \cong \mathfrak{t}/\Lambda$ cited in Lecture 6, and it uses nothing from that lecture.

7.2 Characters and cocharacters

Definition 7.4. A *character* of a torus T is a continuous homomorphism $\chi\colon T \to \mathrm{U}(1)$. A *cocharacter* is a continuous homomorphism $\lambda\colon \mathrm{U}(1) \to T$. Under pointwise multiplication, the characters form a group $X^*(T)$ and the cocharacters form a group $X_*(T)$.

These characters are one-dimensional. Trace characters return in Lecture 9.

Theorem 7.5 (Torus character criterion). *Let* T *be a torus with* $\Lambda = \ker\exp$, *and let* $\ell \in \mathfrak{t}^*$. *There is a character* χ_ℓ *with*

$$\chi_\ell(\exp H) = e^{i\ell(H)} \qquad (H \in \mathfrak{t})$$

if and only if $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. *Every character equals* χ_ℓ *for exactly one such* ℓ, *and* $d\chi_\ell = i\ell$.

Hence $\ell \mapsto \chi_\ell$ is an isomorphism from $\{\ell \in \mathfrak{t}^* : \ell(\Lambda) \subset 2\pi\mathbb{Z}\}$ onto $X^*(T)$.

Proof. Suppose $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. The homomorphism $H \mapsto e^{i\ell(H)}$ from $\mathfrak{t}$ to $\mathrm{U}(1)$ is constant on cosets of Λ. Since exp is onto with kernel Λ, it defines a homomorphism χ_ℓ on T. Continuity follows as for f above. Conversely, if χ_ℓ exists, then $1 = \chi_\ell(\exp H) = e^{i\ell(H)}$ for $H \in \Lambda$.

Now let χ be any character. It is smooth (Lecture 3). Its differential maps $\mathfrak{t}$ to $\operatorname{Lie}\mathrm{U}(1) = i\mathbb{R}$, so $d\chi = i\ell$ for some $\ell \in \mathfrak{t}^*$. Naturality of exp (Lecture 2) gives

$$\chi(\exp H) = \exp\bigl(d\chi(H)\bigr) = e^{i\ell(H)}.$$

For $H \in \Lambda$ the left side is $\chi(1) = 1$, so $\ell(H) \in 2\pi\mathbb{Z}$. Thus $\chi = \chi_\ell$. The form ℓ is unique, because $i\ell(H)$ is the derivative of $s \mapsto \chi(\exp sH)$ at $s = 0$. Finally, $\chi_\ell\chi_{\ell'} = \chi_{\ell+\ell'}$ by the formula. ∎

For $H \in \Lambda$ define the cocharacter

$$\lambda_H(e^{i\theta}) = \exp\Bigl(\frac{\theta}{2\pi}H\Bigr).$$

Replacing θ by $\theta+2\pi$ multiplies the right side by $\exp H = 1$, so λ_H is well defined. Exercise 7.1 shows that $H \mapsto \lambda_H$ is an isomorphism $\Lambda \to X_*(T)$. Since $\chi_\ell(\lambda_H(e^{i\theta})) = e^{i\theta\ell(H)/2\pi}$,

$$\chi_\ell\bigl(\lambda_H(z)\bigr) = z^{\langle \ell, H\rangle}, \qquad \langle \ell, H\rangle = \frac{\ell(H)}{2\pi} \in \mathbb{Z}.$$

So $X^*(T)$ is the dual lattice of $X_*(T) \cong \Lambda$. In crystal physics terms, $X^*(T)$ is the reciprocal lattice of Λ, in the convention $k \cdot R \in 2\pi\mathbb{Z}$, and χ_ℓ is a plane wave that is periodic on Λ. From now on we identify $X^*(T)$ with $\{\ell : \ell(\Lambda) \subset 2\pi\mathbb{Z}\}$ and $X_*(T)$ with Λ.

A finite-dimensional representation of T is unitary for an averaged inner product (Lecture 5). The operators of the representation commute and are unitary, so they diagonalise simultaneously. Hence the representation space is a direct sum of lines on which $\exp H$ acts by $e^{i\ell(H)}$. These $\ell \in X^*(T)$ are the *weights* of the representation.

Example 7.6. Let $T \subset \mathrm{U}(n)$ be the diagonal torus. Then $\mathfrak{t} = \{\operatorname{diag}(i\theta_1, \dots, i\theta_n) : \theta \in \mathbb{R}^n\}$ and $\Lambda = \{2\pi i \operatorname{diag}(k_1, \dots, k_n) : k \in \mathbb{Z}^n\}$. Put $\ell_m(\operatorname{diag}(i\theta)) = \sum_j m_j\theta_j$. Then $\ell_m \in X^*(T)$ if and only if $m \in \mathbb{Z}^n$, and

$$\chi_{\ell_m}\bigl(\operatorname{diag}(z_1, \dots, z_n)\bigr) = z_1^{m_1} \cdots z_n^{m_n}.$$

For $H = 2\pi i \operatorname{diag}(k)$ we get $\lambda_H(z) = \operatorname{diag}(z^{k_1}, \dots, z^{k_n})$ and $\langle \ell_m, H\rangle = \sum_j m_j k_j$.

7.3 Roots, coroots and the centre

Let K be a compact connected Lie group with maximal torus T (Lecture 6). We use the set $R \subset \mathfrak{t}^*$ of real roots from Lecture 6, and we import the root data. A root α has the root space $\mathfrak{g}_\alpha = \{X \in \mathfrak{k}_{\mathbb{C}} : [H, X] = i\alpha(H)X \text{ for } H \in \mathfrak{t}\}$, and

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \qquad \operatorname{Ad}(\exp H)X = e^{i\alpha(H)}X \quad (X \in \mathfrak{g}_\alpha,\ H \in \mathfrak{t}).$$

Each root α has a coroot H_α. It is the unique element

$$H_\alpha \in \mathfrak{t} \cap [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \qquad \text{with} \qquad \alpha(H_\alpha) = 2.$$

The condition $\alpha(H) = 2$ alone does not fix H once $\dim \mathfrak{t} \geq 2$, because then $\ker \alpha \neq 0$. There is a homomorphism $\varphi_\alpha \colon \mathrm{SU}(2) \to K$ with $d\varphi_\alpha(H_0) = H_\alpha$, where $H_0 = \mathrm{diag}(i, -i)$ is the matrix A of Lecture 1 and $-2X_3$ of Lecture 4. Its differential is the compact form of a root $\mathfrak{sl}_2$-triple (H, Chapter 11). It integrates because $\mathrm{SU}(2)$ is simply connected (Lecture 4).

The Lie Algebras course works with complex-linear forms. Its coroot h_α spans the line $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$, and $\mathfrak{t} \cap \mathbb{C}h_\alpha = \mathbb{R}H_\alpha$. The following table compares the two conventions, for $H \in \mathfrak{t}$.

Object	Lie Algebras course	Lectures 6 to 9
root	complex-linear $\mu = i\alpha$ on $\mathfrak{t}_\mathbb{C}$	real α on $\mathfrak{t}$
eigenvalue of $\mathrm{ad}\, H$ on $\mathfrak{g}_\alpha$	$\mu(H)$	$i\alpha(H)$
coroot	h_α, with $\mu(h_\alpha) = 2$	$H_\alpha = ih_\alpha$, with $\alpha(H_\alpha) = 2$

Definition 7.7. Define subgroups Q and P of $\mathfrak{t}^*$, and $Q^\vee$ and $P^\vee$ of $\mathfrak{t}$, by

$$Q = \mathbb{Z}R, \qquad P = \{\ell \in \mathfrak{t}^* : \ell(H_\alpha) \in \mathbb{Z} \text{ for all } \alpha\},$$
$$Q^\vee = \mathbb{Z}\{2\pi H_\alpha : \alpha \in R\}, \quad P^\vee = \{H \in \mathfrak{t} : \alpha(H) \in 2\pi\mathbb{Z} \text{ for all } \alpha\}.$$

A real form $\ell \in \mathfrak{t}^*$ is the complex weight $\mu = i\ell$ of the Lie Algebras course. With $h_\alpha = -iH_\alpha$ we get $\mu(h_\alpha) = \ell(H_\alpha)$. When $\mathfrak{k}$ is semisimple, P is the weight lattice of that course, written in real form.

The factors 2π in $Q^\vee$ and $P^\vee$ fit the convention $\Lambda = \ker \exp$, in which characters satisfy $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. Some texts use $\{H : \exp(2\pi H) = 1\}$ in place of Λ, and write $\chi(\exp 2\pi H) = e^{2\pi i \ell(H)}$. This divides the lattices Λ, $Q^\vee$ and $P^\vee$ in $\mathfrak{t}$ by 2π. The weight lattices $X^*(T)$, Q and P are unchanged.

For example, take $K = \mathrm{SU}(2)$, with $\mathfrak{t} = \mathbb{R}H_0$. Since $\exp(sH_0) = \mathrm{diag}(e^{is}, e^{-is})$, we have $\Lambda = 2\pi\mathbb{Z}H_0$. Since $[H_0, E_{12}] = 2iE_{12}$, the positive root has $\alpha(H_0) = 2$, so $H_\alpha = H_0$. Hence $Q^\vee = 2\pi\mathbb{Z}H_0$ and $P^\vee = \pi\mathbb{Z}H_0$. Identifying ℓ with $m = \ell(H_0)$, we get $Q = 2\mathbb{Z}$ and $P = \mathbb{Z}$.

The covering of Lecture 4 identifies the Lie algebra of $\mathrm{SO}(3)$ with $\mathfrak{su}(2)$. The element sH_0 then exponentiates to the identity of $\mathrm{SO}(3)$ exactly when $\exp(sH_0) = \pm I$ in $\mathrm{SU}(2)$. So $\mathrm{SO}(3)$ has $\Lambda = \pi\mathbb{Z}H_0$. These are the only two lattices between $Q^\vee$ and $P^\vee$. The torus character criterion gives $X^*(T) = \mathbb{Z}$ for $\mathrm{SU}(2)$ and $X^*(T) = 2\mathbb{Z}$ for $\mathrm{SO}(3)$. So the larger kernel lattice has the smaller character lattice. The table collects these values, and Fig. 7.1 draws them.

Lattice	Lives in	Defined by	$\mathrm{SU}(2)$	$\mathrm{SO}(3)$
$Q^\vee$	$\mathfrak{t}$	$\mathbb{Z}\{2\pi H_\alpha\}$	$2\pi\mathbb{Z}H_0$	$2\pi\mathbb{Z}H_0$
Λ	$\mathfrak{t}$	$\ker \exp$	$2\pi\mathbb{Z}H_0$	$\pi\mathbb{Z}H_0$
$P^\vee$	$\mathfrak{t}$	$\alpha(H) \in 2\pi\mathbb{Z}$ for all α	$\pi\mathbb{Z}H_0$	$\pi\mathbb{Z}H_0$
Q	$\mathfrak{t}^*$	$\mathbb{Z}R$	$m \in 2\mathbb{Z}$	$m \in 2\mathbb{Z}$
$X^*(T)$	$\mathfrak{t}^*$	$\ell(\Lambda) \subset 2\pi\mathbb{Z}$	$m \in \mathbb{Z}$	$m \in 2\mathbb{Z}$
P	$\mathfrak{t}^*$	$\ell(H_\alpha) \in \mathbb{Z}$ for all α	$m \in \mathbb{Z}$	$m \in \mathbb{Z}$

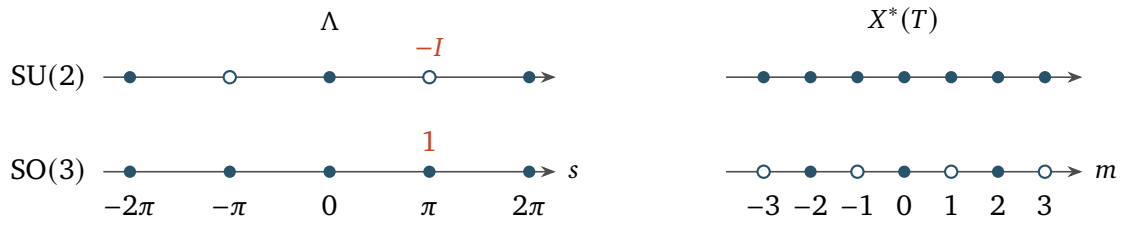


Figure 7.1: The kernel lattice Λ on the left, in the coordinate s of sH_0, and the character lattice $X^*(T)$ on the right, in the coordinate $m = \ell(H_0)$. Filled dots belong to the lattice. A value of m is filled exactly when $ms \in 2\pi\mathbb{Z}$ for every filled s in the same row. The element πH_0 exponentiates to $-I$ in SU(2) and to 1 in SO(3).

Proposition 7.8 (Lattice chain)**.**

$$Q^\vee \subset \Lambda \subset P^\vee, \qquad Q \subset X^*(T) \subset P.$$

Proof. Let $H \in \Lambda$ and $0 \neq X \in \mathfrak{g}_\alpha$. Then $X = \mathrm{Ad}(\exp H)X = e^{i\alpha(H)}X$, so $\alpha(H) \in 2\pi\mathbb{Z}$. Hence $\Lambda \subset P^\vee$. The same computation shows $\alpha \in X^*(T)$, so $Q \subset X^*(T)$. Next, naturality of exp gives

$$\exp(2\pi H_\alpha) = \varphi_\alpha\big(\exp(2\pi H_0)\big) = \varphi_\alpha\begin{pmatrix} e^{2\pi i} & 0 \\ 0 & e^{-2\pi i}\end{pmatrix} = \varphi_\alpha(I) = 1.$$

So $Q^\vee \subset \Lambda$. Finally, if $\ell \in X^*(T)$, then $\ell(2\pi H_\alpha) \in 2\pi\mathbb{Z}$, so $\ell(H_\alpha) \in \mathbb{Z}$ and $\ell \in P$. ∎

Theorem 7.9 (Centre)**.** *Let K be compact connected with maximal torus T. Then $Z(K) \subset T$, and* exp *restricts to a surjective homomorphism $P^\vee \to Z(K)$ with kernel Λ. Thus $Z(K) \cong P^\vee/\Lambda$.*

Proof. Let $z \in Z(K)$. By Lecture 6, $z = gtg^{-1}$ for some $g \in K$ and $t \in T$. Since z is central, $t = g^{-1}zg = z$, so $z \in T$.

Next we show that $g \in K$ is central if and only if $\mathrm{Ad}(g) = \mathrm{id}$. If g is central, differentiating $g\exp(sX)g^{-1} = \exp(sX)$ at $s = 0$ gives $\mathrm{Ad}(g) = \mathrm{id}$. If $\mathrm{Ad}(g) = \mathrm{id}$, then $g\exp(X)g^{-1} = \exp(\mathrm{Ad}(g)X) = \exp X$. The elements $\exp X$ generate the connected group K (Lecture 2), so g is central.

Now write $z = \exp H$ with $H \in \mathfrak{t}$, using the surjectivity of exp in the torus classification, Theorem 7.3. The operator $\mathrm{Ad}(\exp H) = e^{\mathrm{ad}\,H}$ is the identity on $\mathfrak{t}_\mathbb{C}$ and multiplication by $e^{i\alpha(H)}$ on $\mathfrak{g}_\alpha$. Hence, by the criterion just proved, $\exp H$ is central exactly when $H \in P^\vee$. The kernel of exp on $P^\vee$ is $\Lambda \cap P^\vee = \Lambda$. ∎

For the fundamental group we cite the following theorem.

Theorem 7.10 (Fundamental group; H, Chapter 13; BD, Section V.7)**.** *Let K be a compact connected Lie group with maximal torus T. Then*

$$\pi_1(K) \cong \Lambda/Q^\vee = X_*(T)/Q^\vee.$$

Call K *semisimple* if $\mathfrak{k}$ is semisimple. Then R spans $\mathfrak{t}^*$, and the simple roots form a basis of $\mathfrak{t}^*$. The simple coroots form a basis of $\mathfrak{t}$ (the Lie Algebras course, via $h_\alpha = -iH_\alpha$). So all four

subgroups are lattices, and the inclusions above have finite index. Weyl's theorem states that a compact connected semisimple group has finite fundamental group and compact universal cover (H, Chapter 13; Knapp, Chapter IV).

7.4 Global forms

Semisimple groups

By Weyl's theorem, the universal cover of our compact connected semisimple K is compact. Let K_{sc} be a compact, connected, simply connected, semisimple Lie group with Lie algebra $\mathfrak{k}$ and maximal torus T_{sc}. The fundamental group theorem, Theorem 7.10, gives $\Lambda_{sc} = Q^\vee$. The torus character criterion and the centre theorem, Theorems 7.5 and 7.9, then give

$$X^*(T_{sc}) = P, \qquad Z(K_{sc}) \cong P^\vee/Q^\vee.$$

Proposition 7.11 (Forms of a semisimple algebra). *(a) Every connected Lie group with Lie algebra $\mathfrak{k}$ is isomorphic to K_{sc}/C for a subgroup $C \subset Z(K_{sc})$.*

(b) For $K = K_{sc}/C$, the image T of T_{sc} is a maximal torus of K, with kernel lattice

$$\Lambda_C = \{H \in \mathfrak{t} : \exp_{sc} H \in C\}, \qquad \Lambda_C/Q^\vee \cong C.$$

The map $C \mapsto \Lambda_C$ is a bijection from subgroups of $Z(K_{sc})$ to lattices Λ with $Q^\vee \subset \Lambda \subset P^\vee$.

(c) $X^*(T) = \{\ell \in P : \chi_\ell(c) = 1 \text{ for all } c \in C\}$.

Proof. (a) By Lecture 4, such a group is isomorphic to K_{sc}/C with C discrete and normal. Such a subgroup C is central.

(b) First, T is a maximal torus with kernel lattice Λ_C. The covering map $p\colon K_{sc} \to K$ is a local isomorphism. So $\operatorname{Lie} T = \mathfrak{t}$ and $\exp_K = p \circ \exp_{sc}$. Hence $\exp_K H = 1$ exactly when $\exp_{sc} H \in C$. A larger torus would have a larger abelian Lie algebra, but $\mathfrak{t}$ is maximal abelian in $\mathfrak{k}$ (Lecture 6). So T is maximal.

Next, we prove $Q^\vee \subset \Lambda_C \subset P^\vee$ and $\Lambda_C/Q^\vee \cong C$. Since $C \subset Z(K_{sc})$, the centre theorem, Theorem 7.9, gives $Q^\vee \subset \Lambda_C \subset P^\vee$. The map $\exp_{sc}\colon \Lambda_C \to C$ is onto, since $C \subset T_{sc} = \exp_{sc}(\mathfrak{t})$. Its kernel is $Q^\vee$, so $\Lambda_C/Q^\vee \cong C$.

Conversely, let $Q^\vee \subset \Lambda \subset P^\vee$, and put $C = \exp_{sc}(\Lambda)$. By the centre theorem, Theorem 7.9, C is a subgroup of $Z(K_{sc})$. Clearly $\Lambda \subset \Lambda_C$. For the reverse inclusion, let $H \in \Lambda_C$. Then $\exp_{sc} H \in C = \exp_{sc}(\Lambda)$, so $\exp_{sc} H = \exp_{sc} H'$ for some $H' \in \Lambda$. Hence $H - H'$ lies in the kernel of $\exp_{sc}$ on $\mathfrak{t}$, which is $Q^\vee \subset \Lambda$. So $H \in \Lambda$, and $\Lambda_C = \Lambda$. We also showed $\exp_{sc}(\Lambda_C) = C$ for every C. So $C \mapsto \Lambda_C$ and $\Lambda \mapsto \exp_{sc}(\Lambda)$ are inverse to each other.

(c) A character of $T = T_{sc}/C$ is a character of T_{sc} that is trivial on C. ∎

Remark 7.12. Distinct lattices can give isomorphic groups. Let σ be an automorphism of $\mathfrak{k}$ with $\sigma(\mathfrak{t}) = \mathfrak{t}$. It integrates to an automorphism Σ of K_{sc} (Lecture 4). Since $\Sigma(\exp H) = \exp(\sigma H)$, we get $\Sigma(Z(K_{sc})) = Z(K_{sc})$ and $\Lambda_{\Sigma(C)} = \sigma(\Lambda_C)$. Hence $K_{sc}/C \cong K_{sc}/\Sigma(C)$. Exercise 7.5 gives an example. We do not discuss the converse.

The *simply connected form* is $C = 1$, with $\Lambda = Q^\vee$ and $X^*(T) = P$. The *adjoint form* is $K_{ad} = K_{sc}/Z(K_{sc})$, with $\Lambda = P^\vee$. Its centre $P^\vee/P^\vee$ is trivial by the centre theorem, Theorem 7.9. Write T_{ad} for the image of T_{sc} in K_{ad}. Every root is an integer combination of the simple roots α_i (the Lie Algebras course). So the lattice $P^\vee$ is defined by the conditions $\alpha_i(H) \in 2\pi\mathbb{Z}$ for the simple roots alone. Since the simple roots form a basis of $\mathfrak{t}^*$, we get $X^*(T_{ad}) = \mathbb{Z}\{\alpha_i\} = Q$.

Proposition 7.13 (Descent by central character)**.** *Let V be the irreducible representation of K_{sc} with dominant highest weight $\lambda \in P$, and let $C \subset Z(K_{sc})$. Then V descends to K_{sc}/C if and only if*

$$\chi_\lambda(c) = 1 \qquad \text{for all } c \in C.$$

The representation V exists by the highest-weight theorem of the Lie Algebras course and Lecture 4.

Proof. By Schur's lemma, a central element c acts on V by a scalar. On a highest weight vector this scalar is $\chi_\lambda(c)$. By Lecture 4, V descends to K_{sc}/C if and only if C acts trivially. ∎

Groups with a central torus

In U(2) the element $-I$ is both a scalar and an element of SU(2). A representation must act on it consistently. Exercise 4.3 did this by hand. We now state the condition for a general central torus.

If $\mathfrak{k}$ has nonzero centre, then $Z(K)$ is infinite and R does not span $\mathfrak{t}^*$. Moreover, $P^\vee$ contains the centre of $\mathfrak{k}$, so $P^\vee$ is not a lattice. The proposition on forms of a semisimple algebra, Proposition 7.11, does not apply. We use a structure theorem instead.

We recall the global structure theorem of Lecture 6, together with the splitting of that lecture.

Theorem 7.14 (Global structure; recalled from Lecture 6; BD, Section V.8; Knapp, Chapter IV)**.** *Let K be compact connected. Let T_0 be the identity component of $Z(K)$, with Lie algebra $\mathfrak{t}_0$. Let K_{sc} be the simply connected group with Lie algebra $[\mathfrak{k},\mathfrak{k}]$. Then T_0 is a torus, $\mathfrak{k} = \mathfrak{t}_0 \oplus [\mathfrak{k},\mathfrak{k}]$, the ideal $[\mathfrak{k},\mathfrak{k}]$ is semisimple, and K_{sc} is compact. Let $\varphi\colon K_{sc} \to K$ integrate the inclusion of $[\mathfrak{k},\mathfrak{k}]$. Then*

$$T_0 \times K_{sc} \to K, \qquad (t,g) \mapsto t\,\varphi(g),$$

is a surjective homomorphism with finite kernel F. Thus $K \cong (T_0 \times K_{sc})/F$.

Proposition 7.15. *In the global structure theorem, Theorem 7.14, let T_{sc} be a maximal torus of K_{sc} with Lie algebra $\mathfrak{t}_{sc}$, and let T be the image of $T_0 \times T_{sc}$. Identify $\mathfrak{t}^* = \mathfrak{t}_0^* \oplus \mathfrak{t}_{sc}^*$, and let $P_{sc} \subset \mathfrak{t}_{sc}^*$ be the weight lattice of K_{sc}. Then $T \cong (T_0 \times T_{sc})/F$ is a maximal torus of K, and*

$$X^*(T) = \big\{(\ell_0,\ell) \in X^*(T_0) \times P_{sc} :\ \chi_{\ell_0}(t)\,\chi_\ell(g) = 1 \text{ for all } (t,g) \in F\big\}.$$

Proof. A finite normal subgroup of a connected group is central (Lecture 4). So $F \subset T_0 \times Z(K_{sc}) \subset T_0 \times T_{sc}$ by the centre theorem, Theorem 7.9. Hence $T \cong (T_0 \times T_{sc})/F$. Its Lie algebra $\mathfrak{t}_0 \oplus \mathfrak{t}_{sc}$ is maximal abelian in $\mathfrak{k}$, because $\mathfrak{t}_{sc}$ is maximal abelian in $[\mathfrak{k},\mathfrak{k}]$. So T is a maximal torus.

Each root space satisfies $\mathfrak{g}_\alpha = [\mathfrak{t}, \mathfrak{g}_\alpha] \subset [\mathfrak{k}, \mathfrak{k}]_{\mathbb{C}}$. So $H_\alpha \in \mathfrak{t} \cap [\mathfrak{k}, \mathfrak{k}] = \mathfrak{t}_{sc}$, and the coroots of K are those of K_{sc}. Thus $Q^\vee \subset \mathfrak{t}_{sc}$ is the kernel lattice of T_{sc}. Put

$$\Lambda_0 = \ker\bigl(\exp|_{\mathfrak{t}_0}\bigr) = \Lambda \cap \mathfrak{t}_0.$$

The torus $T_0 \times T_{sc}$ has kernel lattice $\Lambda_0 \oplus Q^\vee$. By the torus character criterion, Theorem 7.5, its characters are the pairs with $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$ and $\ell \in P_{sc}$. The characters of T are those trivial on F. ∎

In other words, (ℓ_0, ℓ) is a character of T exactly when $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$, $\ell \in P_{sc}$, and the product $\chi_{\ell_0}\chi_\ell$ is trivial on F.

The group F is determined by Λ. The map $T_0 \times T_{sc} \to T$ has differential the identity of $\mathfrak{t} = \mathfrak{t}_0 \oplus \mathfrak{t}_{sc}$. By the torus classification, Theorem 7.3,

$$T_0 \times T_{sc} \cong \mathfrak{t}/(\Lambda_0 \oplus Q^\vee), \qquad T \cong \mathfrak{t}/\Lambda.$$

The map is induced by the identity of $\mathfrak{t}$, so its kernel is

$$F \cong \Lambda/(\Lambda_0 \oplus Q^\vee).$$

7.5 Examples

SU(2) and SO(3)

We use the values found after the definition of the lattices, and drawn in Fig. 7.1. For SU(2) we have $Q^\vee = \Lambda$. With $m = \ell(H_0)$,

$$P = \mathbb{Z}, \qquad Q = 2\mathbb{Z}, \qquad P^\vee = \pi\mathbb{Z}H_0, \qquad Z(\mathrm{SU}(2)) = \exp(\pi\mathbb{Z}H_0) = \{\pm I\}.$$

The adjoint form is $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ (Lecture 4). It has $\Lambda = \pi\mathbb{Z}H_0$ and

$$X^*(T) = \{m : m\pi \in 2\pi\mathbb{Z}\} = 2\mathbb{Z} = Q, \qquad \pi_1(\mathrm{SO}(3)) \cong \pi\mathbb{Z}/2\pi\mathbb{Z} \cong \mathbb{Z}/2.$$

The module V_m has dimension $m+1$ and weights $m, m-2, \dots, -m$. Physicists call it the spin-$m/2$ representation. Since $-I = \exp(\pi H_0)$, the element $-I$ acts on V_m by $e^{i\pi m} = (-1)^m$. So V_m descends to SO(3) exactly when m is even.

SU(3) and PSU(3)

For the diagonal torus of SU(3),

$$\mathfrak{t} = \Bigl\{\mathrm{diag}(i\theta_1, i\theta_2, i\theta_3) : \sum_j \theta_j = 0\Bigr\}, \qquad \Lambda = \Bigl\{2\pi i\,\mathrm{diag}(k) : k \in \mathbb{Z}^3,\ \sum_j k_j = 0\Bigr\}.$$

The roots are $\alpha_{jk}(H) = \theta_j - \theta_k$ for $j \neq k$, with coroots $H_{jk} = i(E_{jj} - E_{kk})$. Indeed $[E_{jk}, E_{kj}] = E_{jj} - E_{kk}$ and $\alpha_{jk}(H_{jk}) = 2$. Hence $Q^\vee = \Lambda$. Write $\ell_m(H) = \sum_j m_j\theta_j$ with $m \in \mathbb{R}^3$. Adding a multiple of $(1,1,1)$ to m does not change ℓ_m, since $\sum_j \theta_j = 0$. The condition $\ell_m(\Lambda) \subset 2\pi\mathbb{Z}$ means $m_j - m_k \in \mathbb{Z}$, and this is also the condition for P.

Such a form has a representative $m \in \mathbb{Z}^3$, obtained by subtracting $m_3(1,1,1)$. Two integer representatives differ by an element of $\mathbb{Z}(1,1,1)$, which changes $m_1+m_2+m_3$ by a multiple of 3. Using integer representatives,

$$X^*(T) = P \cong \mathbb{Z}^3/\mathbb{Z}(1,1,1),$$
$$Q = \{[m] : m \in \mathbb{Z}^3,\ m_1+m_2+m_3 \equiv 0 \bmod 3\}, \qquad P/Q \cong \mathbb{Z}/3.$$

An element of $P^\vee$ has $\theta_j = \theta + 2\pi k_j$ with $3\theta \in 2\pi\mathbb{Z}$. So $Z(\mathrm{SU}(3)) = \{I, \omega I, \omega^2 I\}$ with $\omega = e^{2\pi i/3}$. The adjoint form $\mathrm{PSU}(3) = \mathrm{SU}(3)/Z(\mathrm{SU}(3))$ has $X^*(T) = Q$, trivial centre and $\pi_1 \cong P^\vee/Q^\vee \cong \mathbb{Z}/3$.

Use the simple roots α_{12}, α_{23}, with simple coroots H_{12}, H_{23}. The fundamental weights are then $\varpi_1 = \ell_{(1,0,0)}$ and $\varpi_2 = \ell_{(1,1,0)}$. So the highest weight $a\varpi_1 + b\varpi_2$ is $\ell_{(a+b,b,0)}$. Write $V(a,b)$ for the irreducible module with this highest weight. Take

$$H_\omega = \mathrm{diag}\Big(\frac{2\pi i}{3}, \frac{2\pi i}{3}, -\frac{4\pi i}{3}\Big), \qquad \exp H_\omega = \mathrm{diag}(\omega, \omega, \omega^{-2}) = \omega I.$$

Then $\ell_{(a+b,b,0)}(H_\omega) = 2\pi(a+2b)/3$, so ωI acts on $V(a,b)$ by ω^{a+2b}. Thus $V(a,b)$ descends to PSU(3) if and only if $a \equiv b \bmod 3$. The adjoint module $V(1,1)$ descends. The module $\mathbb{C}^3 = V(1,0)$ does not. In physics terms, the number $a-b$ modulo 3 is the triality, the module $V(1,1)$ is the octet and $V(1,0)$ is the quark triplet.

Figure 7.2 draws the same data in the plane $\mathfrak{t}$ and in the dual plane $\mathfrak{t}^*$. In $\mathfrak{t}$ the kernel lattice of SU(3) is the smaller lattice $Q^\vee$, and that of PSU(3) is the larger lattice $P^\vee$. In $\mathfrak{t}^*$ the order reverses. The character lattice of SU(3) is the larger lattice P, and that of PSU(3) is the smaller lattice Q. The same three marks appear in both planes.

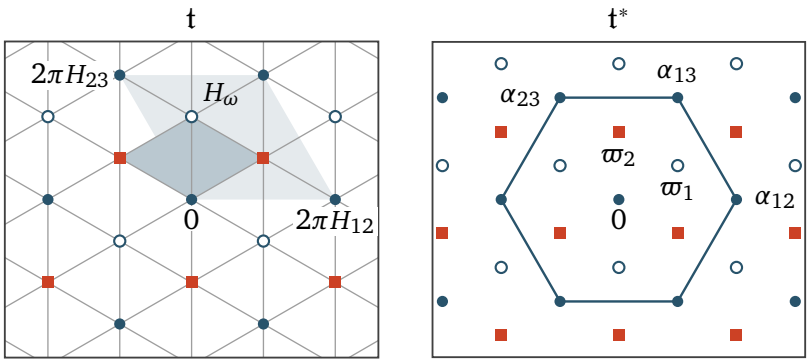


Figure 7.2: The plane $\mathfrak{t}$ of SU(3), with the form $-\mathrm{tr}(XY)$, on the left, and the dual plane $\mathfrak{t}^*$ on the right. Their scales differ. On the left, a disc, circle or square marks a point whose exponential is I, ωI or $\omega^2 I$. Discs form $Q^\vee$, and all marks form $P^\vee$. The thin lines are $\alpha_{jk}(H) \in 2\pi\mathbb{Z}$, and the shaded cells are cells of $Q^\vee$ and of $P^\vee$. On the right, a mark shows the scalar by which ωI acts on a weight: 1, ω or ω^2. Discs form Q, and all marks form P. The hexagon joins the roots, and $\alpha_{13} = \alpha_{12} + \alpha_{23} = \varpi_1 + \varpi_2$.

The group U(2)

Let $T_0 = \{zI : z \in \mathrm{U}(1)\}$ and $K_{sc} = \mathrm{SU}(2)$. The map

$$\Psi\colon\ \mathrm{U}(1) \times \mathrm{SU}(2) \to \mathrm{U}(2), \qquad \Psi(z,g) = zg,$$

is onto. Given u, choose z with $z^2 = \det u$. Then $z^{-1}u \in \mathrm{SU}(2)$. If $zg = I$, then $\det(z^{-1}I) = z^{-2} = 1$, so $z = \pm 1$. Hence

$$\mathrm{U}(2) \cong \big(\mathrm{U}(1) \times \mathrm{SU}(2)\big)/F, \qquad F = \{(1, I), (-1, -I)\}.$$

The lattice statements for semisimple groups fail for U(2) itself. In the semisimple case the roots span $\mathfrak{t}^*$, and Q, P, $Q^\vee$ and $P^\vee$ are lattices. By the computation for diagonal tori, Example 7.6, the diagonal torus of U(2) has $X^*(T) = \mathbb{Z}^2$. The roots are $\pm\alpha$ with $\alpha(\mathrm{diag}(i\theta)) = \theta_1 - \theta_2$, so $Q = \mathbb{Z}(1, -1)$ spans only a line. The set $P^\vee = \{\theta_1 - \theta_2 \in 2\pi\mathbb{Z}\}$ is not a lattice, and $P^\vee/\Lambda \cong \mathrm{U}(1)$ is the group of scalars.

The fundamental group theorem, Theorem 7.10, still applies. Since $[E_{12}, E_{21}] = \mathrm{diag}(1, -1)$, the coroot is $H_\alpha = i\,\mathrm{diag}(1, -1)$, and

$$\pi_1(\mathrm{U}(2)) \cong \mathbb{Z}^2/\mathbb{Z}(1, -1) \cong \mathbb{Z}, \qquad (k_1, k_2) \mapsto k_1 + k_2.$$

Here $\Lambda_0 = 2\pi i\mathbb{Z}I$, and $\Lambda_0 \oplus Q^\vee$ consists of the $2\pi i\,\mathrm{diag}(k_1, k_2)$ with $k_1 \equiv k_2 \bmod 2$. So $\Lambda/(\Lambda_0 \oplus Q^\vee) \cong \mathbb{Z}/2 \cong F$, as predicted.

Now we compute the central compatibility constraint. Recall from Lecture 4 that Π_m denotes the representation of SU(2) on V_m. The element $\mathrm{diag}(w, w^{-1})$, with $w \in \mathrm{U}(1)$, acts on a vector of weight j by w^j. For $k \in \mathbb{Z}$ and $m \geq 0$ define

$$\Pi_{k,m}(z, g) = z^k\,\Pi_m(g) \qquad \text{on } V_m.$$

The representation $\Pi_{k,m}$ is irreducible, already on SU(2). By Lecture 4 it descends to U(2) exactly when it is trivial on F. Since

$$\Pi_{k,m}(-1, -I) = (-1)^k(-1)^m\ \mathrm{id},$$

the constraint is

$$k \equiv m \pmod 2.$$

The descended representation is $\Pi(u) = z^k\Pi_m(z^{-1}u)$ with $z^2 = \det u$. The two choices of z give the same operator exactly because of the constraint. Exercise 7.3 shows that every irreducible representation of U(2) arises in this way.

The description of $X^*(T)$ for a central torus, Proposition 7.15, gives the same constraint. Here k is the exponent on scalars, and j runs over the weights of V_m. Put $\ell_0(isI) = ks$ and $\ell(sH_0) = js$. In T_0, the scalar $z = -1$ is the matrix $-I = \exp(i\pi I)$. So $(-1, -I) = (\exp(i\pi I), \exp(\pi H_0))$, and the pair (ℓ_0, ℓ) is a character of T exactly when $k + j$ is even. Every weight j of V_m satisfies $j \equiv m \bmod 2$, so this condition is the constraint found above.

Assume $k \equiv m \pmod 2$, so that $\Pi_{k,m}$ descends to U(2). Every weight j of V_m satisfies $j \equiv m \pmod 2$, so $k - j$ is even and the exponents below are integers. On $u = \mathrm{diag}(z_1, z_2)$ with $z^2 = z_1z_2$, a vector of weight j is multiplied by

$$z^k\left(\frac{z_1}{z}\right)^j = (z_1z_2)^{(k-j)/2}z_1^{\,j} = z_1^{(k+j)/2}\,z_2^{(k-j)/2}.$$

So $\Pi_{k,m}$ has highest weight $\left(\frac{k+m}{2}, \frac{k-m}{2}\right) \in \mathbb{Z}^2$. For example, $\mathbb{C}^2$ is $(k, m) = (1, 1)$, det is $(2, 0)$, and Ad on the complexification $\mathfrak{su}(2)_\mathbb{C}$ is $(0, 2)$.

Self-checks

1. Does the maximal torus of SO(3) have a character with $\ell(H_0) = 1$?

   Answer. No. Here $\Lambda = \pi\mathbb{Z}H_0$ and $\ell(\pi H_0) = \pi \notin 2\pi\mathbb{Z}$.

2. Compute $Z(\mathrm{PSU}(3))$ and $\pi_1(\mathrm{PSU}(3))$ from the lattices.

   Answer. Here $\Lambda = P^\vee$. So $Z = P^\vee/P^\vee = 1$ and $\pi_1 = P^\vee/Q^\vee \cong \mathbb{Z}/3$.

3. Is there a representation of U(2) on which scalars act by z^2 and SU(2) acts by V_1?

   Answer. No. The element $-I$ is the scalar -1, acting by $(-1)^2 = 1$. The element $-I$ also lies in SU(2), where it acts on V_1 by -1. The two actions disagree.

Summary

A compact connected abelian Lie group is $\mathfrak{t}/\Lambda$, and its characters are the ℓ with $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. For compact connected K, the kernel lattice satisfies $Q^\vee \subset \Lambda \subset P^\vee$. The kernel lattice gives $Z(K) \cong P^\vee/\Lambda$ and, by a cited theorem, $\pi_1(K) \cong \Lambda/Q^\vee$.

For semisimple $\mathfrak{k}$, the connected forms correspond to lattices between $Q^\vee$ and $P^\vee$. Lattices related by an automorphism of $\mathfrak{k}$ preserving $\mathfrak{t}$ give isomorphic groups. With a central torus, write $K \cong (T_0 \times K_{sc})/F$, where $F \cong \Lambda/(\Lambda_0 \oplus Q^\vee)$. A pair (ℓ_0, ℓ) is then a character of the maximal torus exactly when $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$, $\ell \in P_{sc}$, and the product is trivial on F.

Lecture 8 uses $X^*(T)$ to state the classification of representations by highest weights.

Reading

B. Hall, *Lie Groups, Lie Algebras, and Representations* (H), Chapter 13. T. Bröcker and T. tom Dieck, *Representations of Compact Lie Groups* (BD), Sections V.7–8. A. Knapp, *Lie Groups Beyond an Introduction*, Chapter IV. For the discrete-subgroup lemma, Lemma 7.2, see BD, Section I.3, Lemma (3.8).